

Converting Grayscale to Color Using U-Net Image-to-Image Translation.

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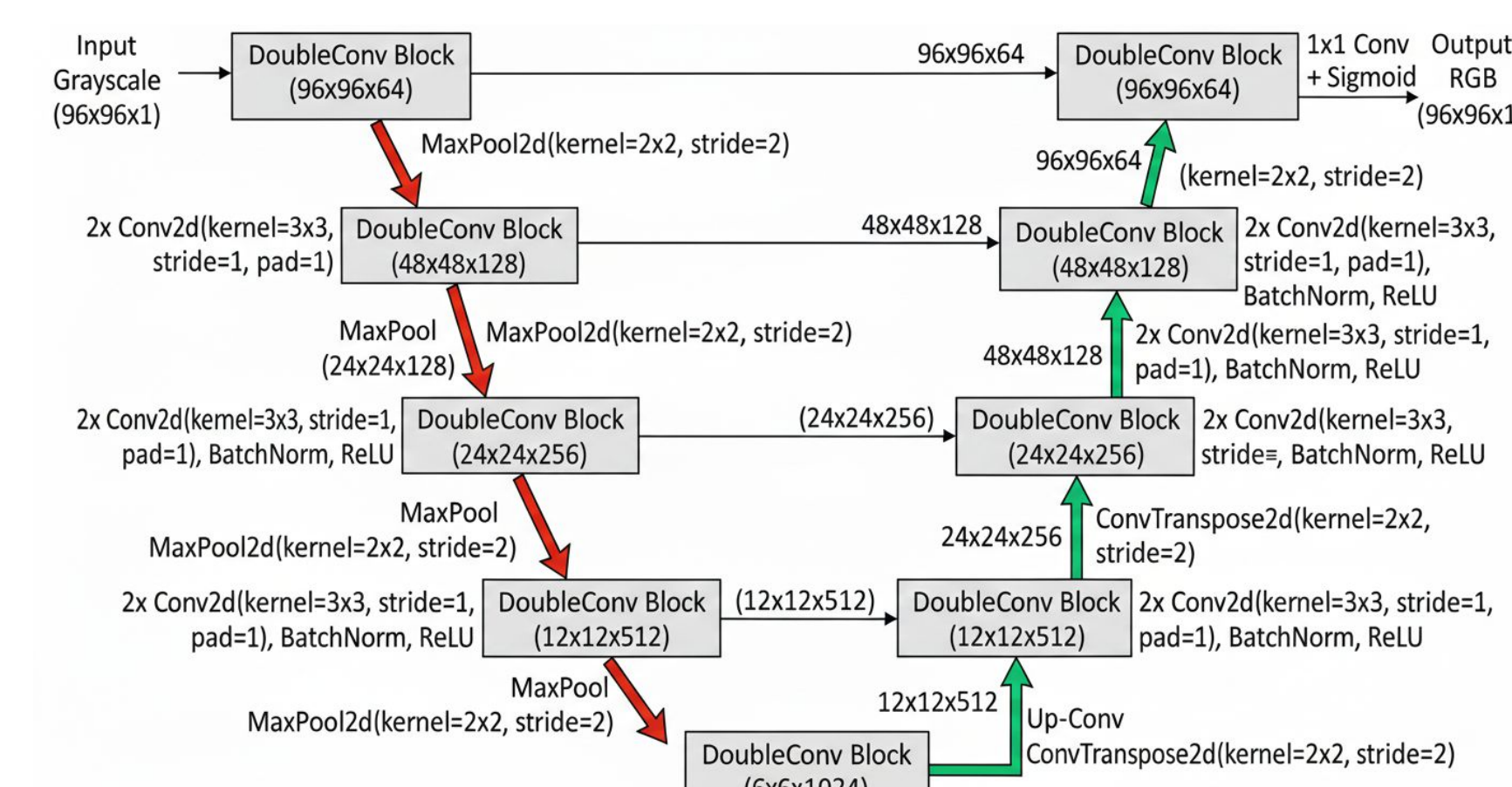
Dataset

- Source of dataset: STL-10
- Number of samples: 10,000 labeled training images and 3,000 test images in total
- Input images are black and white images
- Target images are RGB images paired with an input
- Preprocessing steps: Target images are converted to 1- channel grayscale input images
- Normalization to [0,1]
- Data augmentation includes Random Horizontal Flip



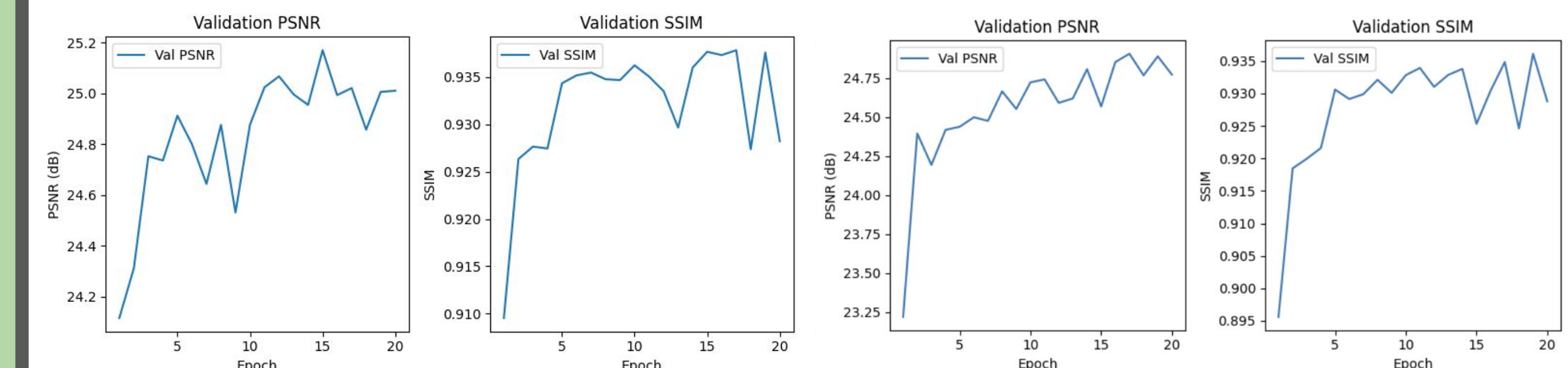
Model Architecture

- U-Net structure
- Double Convolution: 3x3 kernel, stride of 1, pad of 1, BatchNorm, and ReLU
- Max Pooling: 3x3 kernel, stride of 2
- 4 encoding layers, 4 decoding layers
- Bottleneck layer: depth of 4, 6x6 feature maps, 1,024 channels
- Loss function: combination of L1 Loss on output pixels and optional Perceptual Loss
- VGG16 features to produce an RGB image tensor normalized to the [0, 1] range



Training Procedure

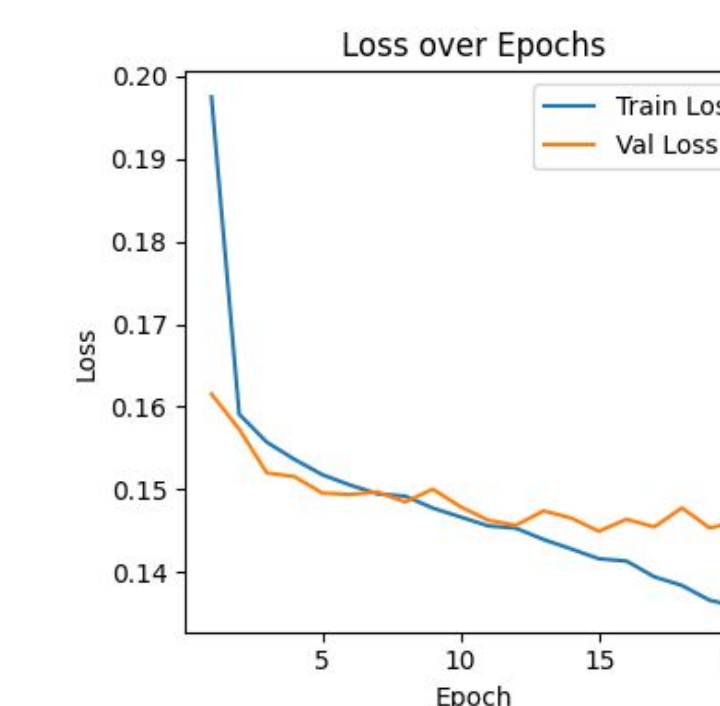
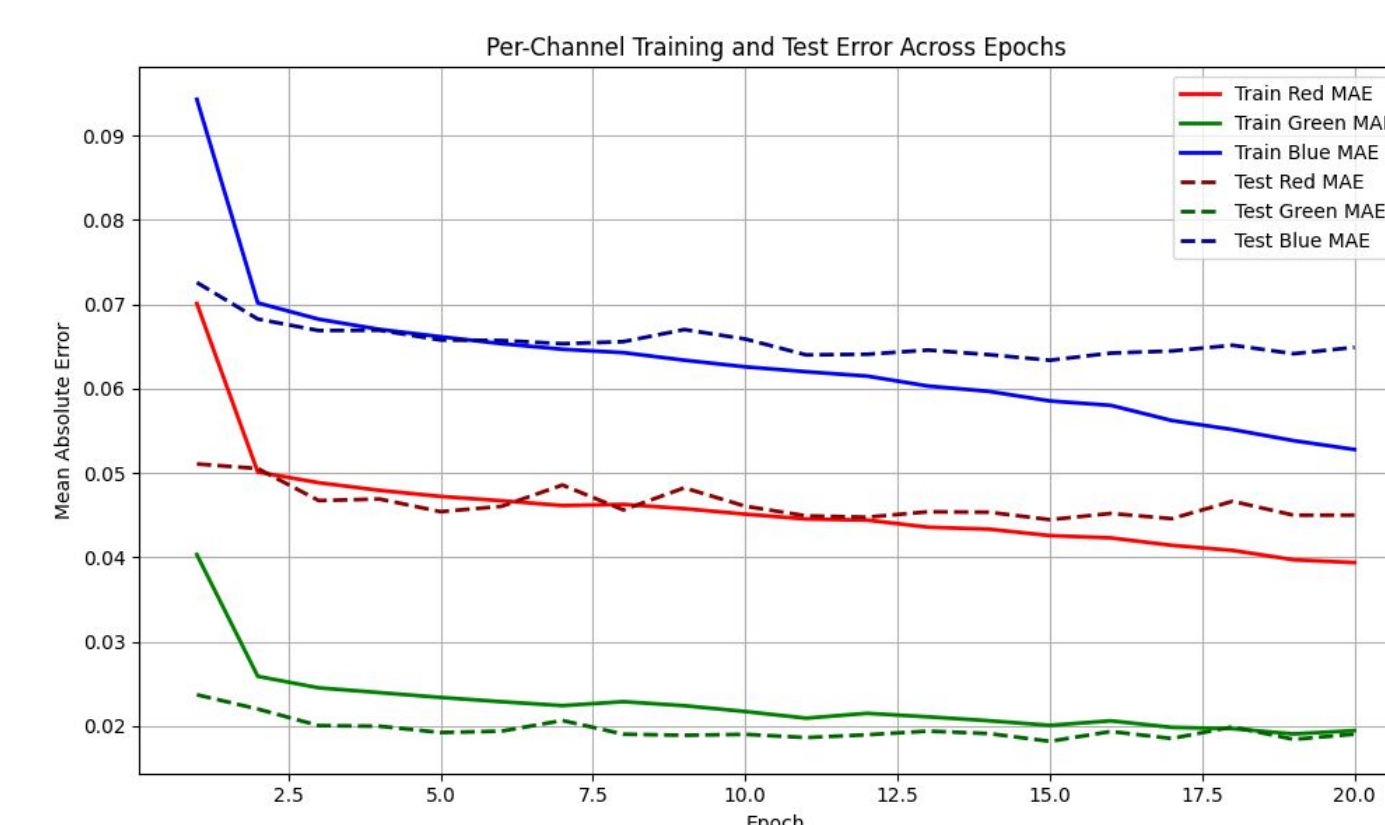
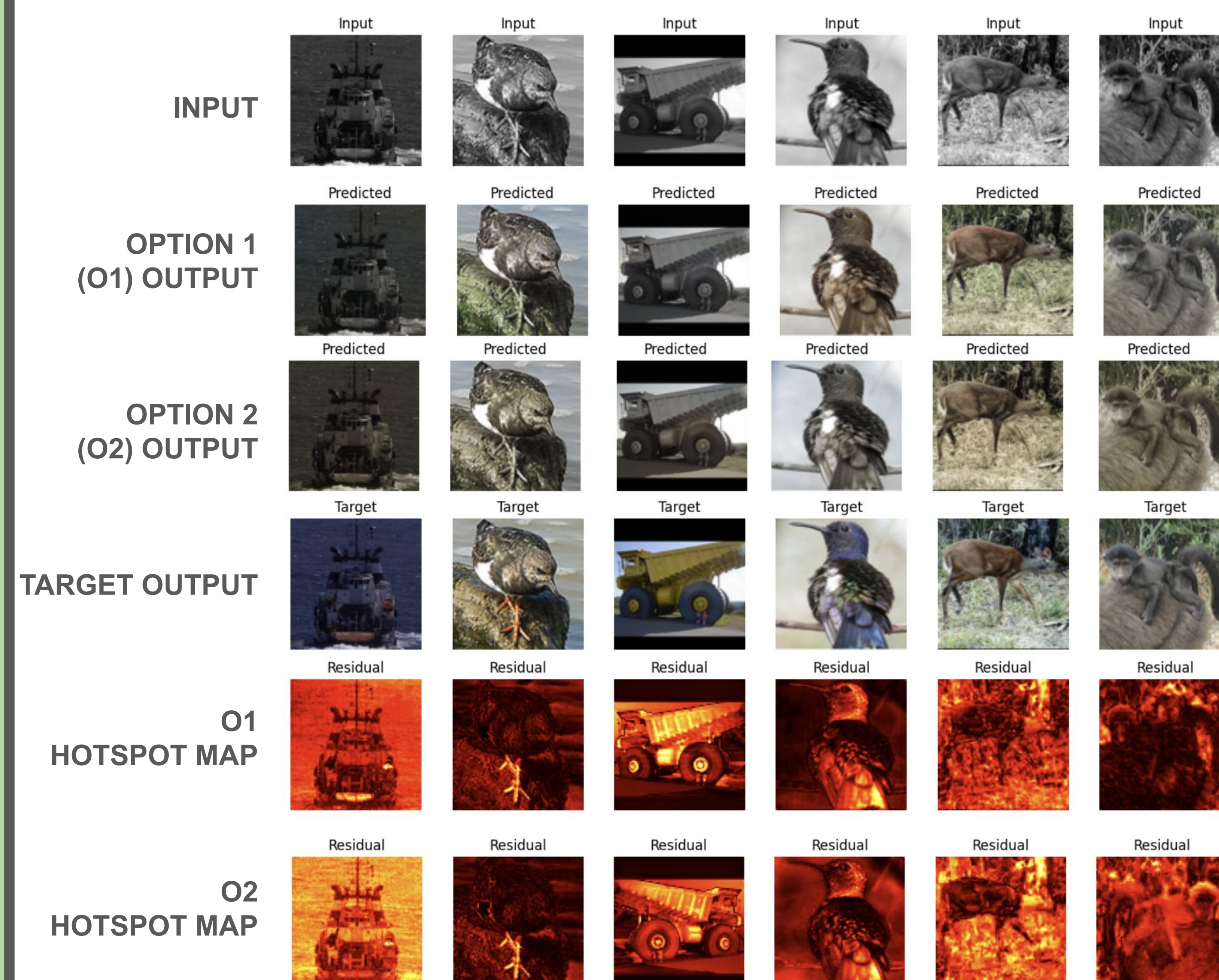
- Option 1 parameters:
 - Seed= 42
 - Batch Size= 64
 - Epochs= 20
 - LR= 0.0002**
 - Weight Decay= 0
 - Loss Type= l1+perceptual**
 - Optimizer = Adam
- Option 2 parameters:
 - Seed= 42
 - Batch Size= 64
 - Epochs= 20,
 - LR= 0.001**
 - Weight Decay= 0
 - Loss Type= l1**
 - Optimizer = Adam
- Augmentation: Geometric
- Runtime type: Google Colab T4 GPU



Option 1

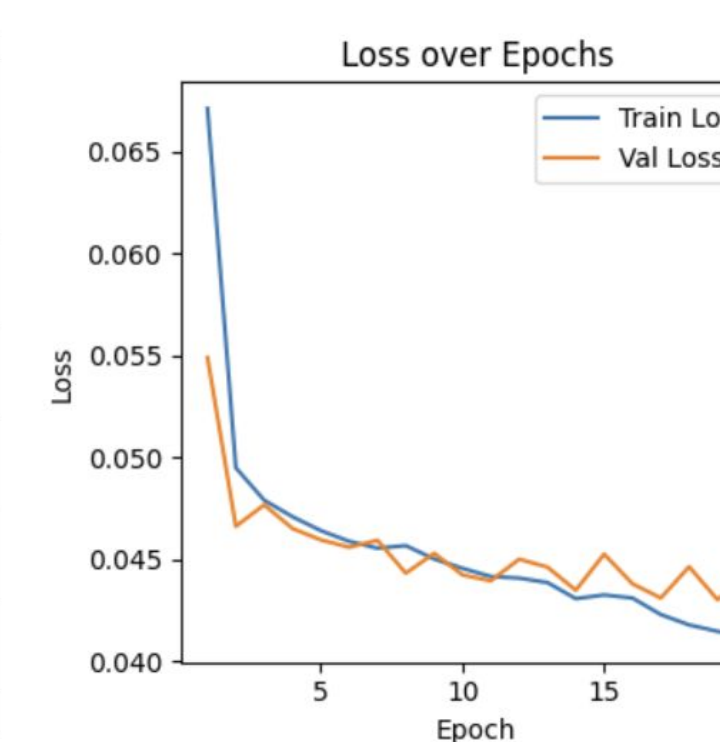
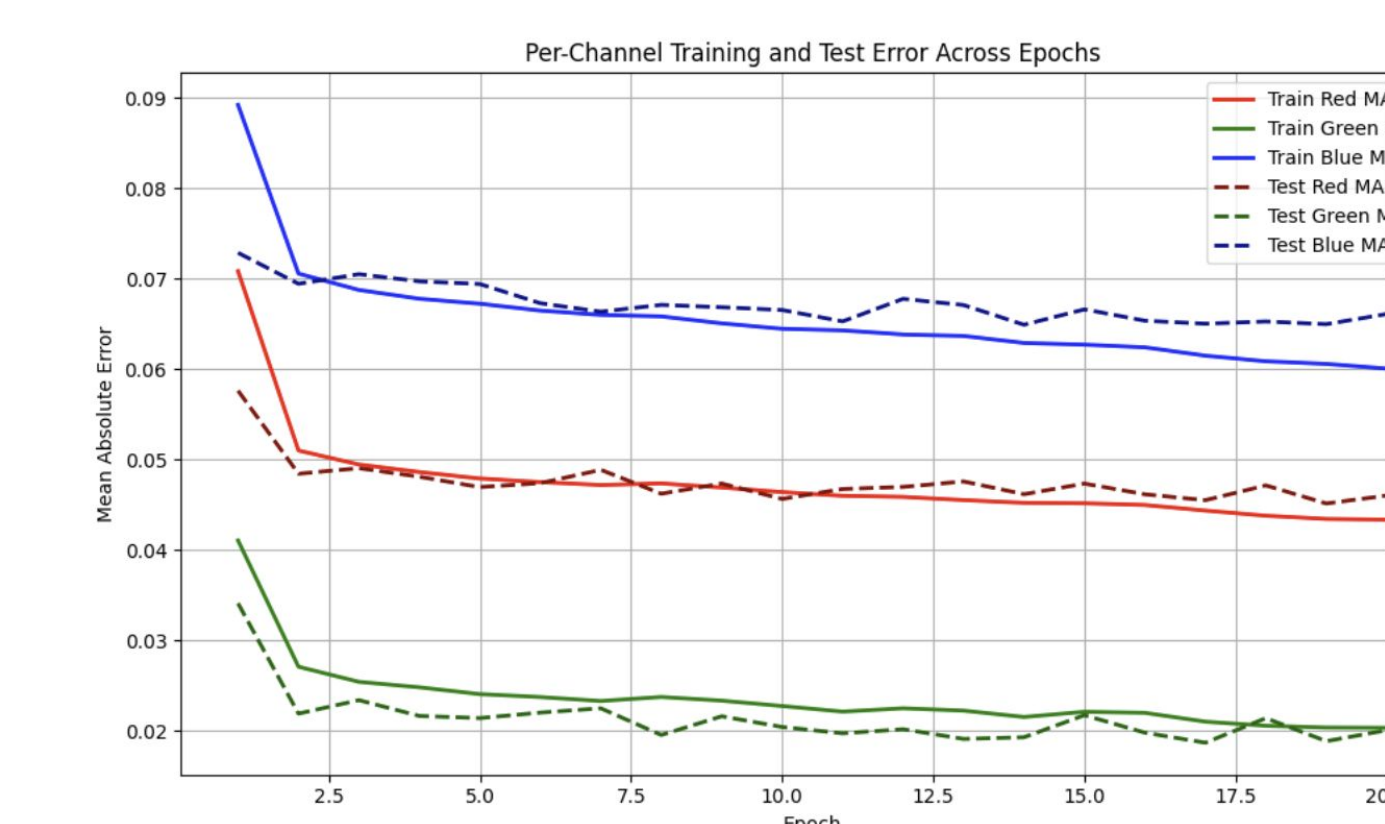
Option 2

Results



Option 1:

- Higher confidence levels
- Color is more saturated but makes more assumptions (ex. deer on green grass, brown bird)
- Higher validation loss
- Residuals hotspot is more focused/localized (bright colors centered, darker background)
- Visually better



Option 2:

- Lower confidence levels
- Color is less saturated and more conservative with assumptions
- Lower validation loss
- Residual hotspot is more widespread (bright colors dispersed, background also contains bright colors)
- Logically better (less loss)

Conclusions & Discussions

- Option 1 (L1 + Perceptual Loss)
 - Works best for strong coloration when accuracy to the original image is not a priority
 - Maximizes coloration
- Option 2 (L1 Loss)
 - Works best when accuracy is a priority
 - Minimizes validation loss
- Observations
 - Features that may vary in color in the real world (e.g. text, a truck bin, bird coloration, etc.) have low confidence coloration and are predicted as grey
 - Features that are very common (e.g. grass) get predicted with higher confidence and are predicted more saturated
- Future considerations:
 - Using Tiny ImageNet to investigate databases with high diversity, but also high colorization challenges